

Turtle Graphics

Turtle graphics were a feature of the Logo programming language designed to help kids learn coding concepts. The feature has since been reproduced in many languages and platforms. Its central idea is an object called a *turtle*.

The turtle acts as a programmable pen that draws lines in a 2D window. Imagine an actual turtle holding a pen on the ground, drawing a line behind it as it moves around. The turtle can adjust the size and color of its pen, or “raise the pen” so that it does not draw as it moves. Turtle programs can produce intricate geometric drawings such as [Figure 9-1](#).

When you put these instructions inside loops and functions, even small programs can create impressive geometric drawings. Consider the following *spiral.py* program:

Python

```
import turtle
turtle.tracer(1, 0) # Makes the turtle draw faster.
for i in range(360):
    turtle.forward(i)
    turtle.left(59)
turtle.exitonclick() # Pause until user clicks in the window.
```

When you run this program, the turtle window opens. The turtle (represented by a triangle) will trace the spiral pattern in [Figure 9-1](#). While not a fractal, it is a beautiful drawing.